

Human motor learning dynamics in high-dimensional tasks: Supporting information

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1 Model

1.1 Human perception model of BoMI

The BoMI mapping

$$\dot{\mathbf{x}} = C\mathbf{u} \quad (1)$$

is described in terms of finger joint velocities $\mathbf{u}$ and cursor velocities $\dot{\mathbf{x}}$. However, we hypothesize that humans interacting with the BoMI perceive these velocities as increments in cursor positions and finger joint angles. Consequently, we adopt a standard technique from adaptive control that allows us to rewrite the BoMI mapping equation using filtered joint and cursor position data [1]. We write the dynamics (Eq (1)) using a mixed frequency-time notation as $s\mathbf{x} = C\mathbf{u}$, where s denotes the Laplace (frequency) variable. Dividing both sides by $s + a$, for some sufficiently large $a > 0$, yields

$$\frac{s}{s+a}\mathbf{x} = -\frac{a}{s+a}\mathbf{x} + \mathbf{x} = \frac{1}{s+a}C\mathbf{u}. \quad (2)$$

Define the signals filtered cursor position $\boldsymbol{\chi} = \mathbf{x}/(s+a)$, and filtered change in hand joint angles $\delta\mathbf{q} = \mathbf{u}/(s+a)$. Equivalently

$$\begin{aligned} \dot{\boldsymbol{\chi}} &= -a\boldsymbol{\chi} + \mathbf{x}, \\ \dot{\delta\mathbf{q}} &= -a\delta\mathbf{q} + \mathbf{u} + \boldsymbol{\xi}_q, \end{aligned} \quad (3)$$

where we add the perceptual noise $\boldsymbol{\xi}_q$ to capture the inaccuracies in the filtering process. Then, in the time-domain (Eq (2)) reduces to $-a\boldsymbol{\chi} + \mathbf{x} = C\delta\mathbf{q}$. Defining $\delta\mathbf{x} = -a\boldsymbol{\chi} + \mathbf{x}$, we get the system equation as

$$\delta\mathbf{x} = C\delta\mathbf{q}. \quad (4)$$

$\delta\mathbf{x}$ and $\delta\mathbf{q}$ are termed *filtered increments in cursor positions*, and *filtered increments in joint angles*, respectively.

1.2 The inverse learning model

We postulate that the participant determines the joint angle velocities through gradient-based learning on a regularized quadratic error function,

$$\dot{\mathbf{u}} = -\eta \nabla_{\mathbf{u}} \left(\frac{1}{2} \left\| \dot{\hat{\mathbf{x}}} - k_P \mathbf{e}_x \right\|^2 + \frac{\mu}{2} \|\mathbf{u}\|^2 \right) + \boldsymbol{\xi}_u, \quad (5)$$

where $\mathbf{u} \in \mathbb{R}^m$, $\hat{\mathbf{x}}$ is the estimated cursor position based on the participant's estimate of mapping matrix $\hat{C}$, $\eta > 0$ is the inverse learning rate, and $\boldsymbol{\xi}_u$ is the exploration noise.

Upon simplification, the gradient in (Eq (5)),

$$\begin{aligned} \nabla_{\mathbf{u}} \left(\frac{1}{2} \left\| \dot{\hat{\mathbf{x}}} - k_P \mathbf{e}_x \right\|^2 + \frac{\mu}{2} \|\mathbf{u}\|^2 \right) &= \nabla_{\mathbf{u}} \left(\frac{1}{2} \left\| \hat{C} \mathbf{u} - k_P \mathbf{e}_x \right\|^2 + \frac{\mu}{2} \|\mathbf{u}\|^2 \right) \\ &= \hat{C}^\top (\hat{C} \mathbf{u} - k_P \mathbf{e}_x) + \mu \mathbf{u} \\ &= (\hat{C}^\top \hat{C} + \mu I) \mathbf{u} - k_P \hat{C}^\top \mathbf{e}_x, \end{aligned} \quad (6)$$

which results in

$$\begin{aligned} \dot{\mathbf{u}} &= -\eta \left((\hat{C}^\top \hat{C} + \mu I) \mathbf{u} - k_P \hat{C}^\top \mathbf{e}_x \right) + \boldsymbol{\xi}_u \\ &= -\eta \left((\Phi^\top \hat{W}^\top \hat{W} \Phi + \mu I) \mathbf{u} - k_P \Phi^\top \hat{W}^\top \mathbf{e}_x \right) + \boldsymbol{\xi}_u. \end{aligned} \quad (7)$$

1.3 Sufficiently rich perceptual noise

Definition 1 (Sufficiently Rich Signal). *A stationary signal $\omega(t)$ is called sufficiently rich of order n if the support of the spectral measure of ω , defined by,*

$$\mathcal{S}_\omega(\theta) = \int_{-\infty}^{\infty} e^{-j\theta\tau} \mathcal{R}_\omega(\tau) d\tau,$$

contains at least n points, where $\mathcal{R}_\omega(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \omega(t) \omega(t + \tau) dt$ is the autocovariance of ω [2].

The perceptual noise $\boldsymbol{\xi}_q$ in the filtered increment in joint angle dynamics (Eq (3)) is a sufficiently rich noise satisfying Definition 1.

Remark 1. *For purposes of simulation and model fitting, $\boldsymbol{\xi}_q$ is treated as white noise with intensity σ_q . $\boldsymbol{\xi}_q$ is a stationary signal as per [2, Definition 5.2.2], with a flat spectral power density $\mathcal{S}_\omega(\theta)$. $\boldsymbol{\xi}_q$ is thus sufficiently rich of any order.*

2 Analysis of adaptive control-based HML model

This section is dedicated to the analysis of the following proposed model of HML from a control-theoretic perspective.

$$\dot{\delta \mathbf{q}} = -a \delta \mathbf{q} + \mathbf{u} + \boldsymbol{\xi}_q \quad (8a)$$

$$\dot{\tilde{C}} = -\gamma \tilde{C} \delta \mathbf{q} \delta \mathbf{q}^\top \quad (8b)$$

$$\dot{\mathbf{e}}_x = -C \mathbf{u} \quad (8c)$$

$$\dot{\mathbf{u}} = -\eta \left((\hat{C}^\top \hat{C} + \mu I) \mathbf{u} - k_P \hat{C}^\top \mathbf{e}_x \right) + \boldsymbol{\xi}_u, \quad (8d)$$

where $\tilde{C} = \hat{C} - C$, and we have used $\dot{\tilde{C}} = \dot{\hat{C}}$. We aim to show that the mathematical model of human forward and inverse learning proposed here, under certain viable assumptions, evolves in a stable fashion and converges to equilibrium exponentially.

The motor learning literature suggests that the forward learning dynamics evolve on a slower timescale than the inverse learning dynamics (refer to the Discussion section in the main text). Consistent with the literature, our parameter fits suggest $\gamma \ll \eta, k_P$, i.e., $\tilde{C}$ dynamics evolve on a slower timescale, and $\mathbf{u}, \mathbf{e}_x$ dynamics evolve on a faster timescale.

Decomposing (Eq (1)) and $\widehat{\delta \mathbf{x}} = \hat{C} \delta \mathbf{q}$ into n equations (one for each cursor motion axis), we obtain $\widehat{\delta \mathbf{x}}_i = \tilde{C}_i \delta \mathbf{q}$ for all $i \in \{1, \dots, n\}$. A similar expression can be written for (Eq (1)). We analyze these equations separately, and for ease of notation, we will drop the index i .

Defining the slow variable $\bar{e}_x = k_P e_x \in \mathbb{R}$, we get the modified reaching error dynamics as

$$\dot{\bar{e}}_x = k_P(\dot{x}^{\text{des}} - \dot{x}) = -k_P C \mathbf{u}. \quad (9)$$

Let $f : \mathbb{R}^m \times \mathbb{R}^{1 \times m} \mapsto \mathbb{R}^{1 \times m}$, $g_1 : \mathbb{R}^m \times \mathbb{R} \times \mathbb{R}^{1 \times m} \mapsto \mathbb{R}^m$, $g_2 : \mathbb{R}^m \mapsto \mathbb{R}$ be defined by

$$\begin{aligned} f(\tilde{C}) &= -\tilde{C} \delta \mathbf{q} \delta \mathbf{q}^\top, \\ g_1(\mathbf{u}, \bar{e}_x, \hat{C}) &= -((\hat{C}^\top \hat{C} + \mu I) \mathbf{u} - \hat{C}^\top \bar{e}_x), \\ g_2(\mathbf{u}) &= -C \mathbf{u}. \end{aligned}$$

Then the HML model dynamics in (Eqs (8b), (8d), (9)) can be equivalently written as

$$\dot{\tilde{C}} = \gamma f(\tilde{C}), \quad (10a)$$

$$\dot{\mathbf{u}} = \eta g_1(\mathbf{u}, \bar{e}_x, \hat{C}) + \boldsymbol{\xi}_u \quad (10b)$$

$$\dot{\bar{e}}_x = k_P g_2(\mathbf{u}), \quad (10c)$$

where we treat $\delta \mathbf{q}$ as an exogenous input for the purpose of analysis. Rewriting system (Eq (10)) in the new timescale $t \mapsto \gamma t$, we get the singularly perturbed system defined by

$$\dot{\tilde{C}} = f(\tilde{C}), \quad (11a)$$

$$\varepsilon \dot{\boldsymbol{\rho}}_u = \varepsilon \begin{bmatrix} \dot{\mathbf{u}} \\ \dot{\bar{e}}_x \end{bmatrix} = \begin{bmatrix} \varepsilon_e g_1(\mathbf{u}, \bar{e}_x, \tilde{C}) + \varepsilon \boldsymbol{\xi}_u / \gamma \\ \varepsilon_u g_2(\mathbf{u}) \end{bmatrix}, \quad (11b)$$

where $\varepsilon = \varepsilon_u \varepsilon_e$, $\varepsilon_u = \gamma / \eta$, and $\varepsilon_e = \gamma / k_P$. $\varepsilon = \gamma^2 / \eta k_P \ll 1$ and thus $\mathbf{u}, \bar{e}_x$ are the fast variables.

It can be verified that under the persistency of excitation of $\delta \mathbf{q}$, $(\mathbf{u}, \bar{e}_x, \tilde{C}) = (0, 0, 0)$ is an isolated equilibrium of system (Eq (11)) in the absence of exploration noise $\boldsymbol{\xi}_u$. Moreover, the functions f, g_1, g_2 are locally Lipschitz, and their partial derivatives up to the second-order are bounded in their respective domains containing the origin.

2.1 Slower-timescale forward learning dynamics

The reduced system associated with (Eq (11)) is

$$\dot{\tilde{C}} = f(\tilde{C}) = -\tilde{C} \delta \mathbf{q} \delta \mathbf{q}^\top, \quad (12)$$

where the fast-varying states are settled at the isolated equilibrium $(\mathbf{u}, \bar{e}_x) = (0, 0)$.

Lemma 1 (*Stability of the Reduced System (Eq (12))*). *The solution of the reduced system (Eq (12)) converges globally exponentially to the origin.*

Proof. $\delta \mathbf{q}$ dynamics with added sufficiently rich noise $\boldsymbol{\xi}_q$ at the equilibrium value of fast-varying states is

$$\begin{aligned} \dot{\delta \mathbf{q}} &= -a\delta \mathbf{q} + \boldsymbol{\xi}_q, \\ \text{or equivalently, } \delta \mathbf{q} &= \frac{1}{s+a}\boldsymbol{\xi}_q. \end{aligned} \quad (13)$$

Invoking [3, Theorem 4.2], with $Q(s) = 1$ and $p(s) = s + a$, $\delta \mathbf{q}$ is persistently exciting. The lemma is then proved using standard adaptive control techniques [2], which leverage the radially unbounded Lyapunov function $V_r(\tilde{C}) = \frac{1}{2}\tilde{C}\tilde{C}^\top$ and persistency of excitation of $\delta \mathbf{q}$. $\square$

2.2 Faster-timescale inverse learning dynamics

Ignoring the noise term $\boldsymbol{\xi}_u$ in $\mathbf{u}$ dynamics, the boundary layer system associated with (Eq (11)) is defined in the new timescale $\tau_u = t/\varepsilon$ as

$$\frac{d\boldsymbol{\rho}_u}{d\tau_u} = \begin{bmatrix} \frac{d\bar{\mathbf{u}}}{d\tau_u} \\ \frac{d\bar{e}_x}{d\tau_u} \end{bmatrix} = \begin{bmatrix} -1/\varepsilon_u \left((\hat{C}^\top \hat{C} + \mu I)\bar{\mathbf{u}} - \hat{C}\bar{e}_x \right) \\ -1/\varepsilon_e (C\bar{\mathbf{u}}) \end{bmatrix}, \quad (14a)$$

where $\mathbf{u}$ is replaced by noise-free joint velocities $\bar{\mathbf{u}}$, and the slow-varying state $\hat{C} = \tilde{C} + C$ is frozen in time.

Lemma 2 (*Stability of the Boundary Layer System (Eq (14))*). *For the boundary layer system (Eq (14)), the equilibrium point at the origin is globally exponentially stable, given that the frozen state $\tilde{C}$ satisfies $\|\tilde{C}\| < \min\{\mu, \theta_c\}$, where $\theta_c \geq 0$ is the largest value that satisfies the cubic inequality $\theta_c^3 - \theta_c^2(2\|C\| + 1) + \theta_c(\|C\|^2 + 2\|C\| + \mu) - \|C\|^2 < 0$.*

Proof. Consider the radially unbounded Lyapunov function

$$\begin{aligned} V_b(\bar{\mathbf{u}}, \bar{e}_x) &= \frac{\varepsilon_u}{2}(\bar{\mathbf{u}} - \mathbf{u}^*)^\top (\bar{\mathbf{u}} - \mathbf{u}^*) + \varepsilon_e \bar{e}_x^2, \\ &= \frac{1}{2}\boldsymbol{\rho}_u^\top P \boldsymbol{\rho}_u, \end{aligned}$$

where $\mathbf{u}^* = \bar{C}^{-1}\hat{C}^\top \bar{e}_x$, $\boldsymbol{\rho}_u = [\bar{\mathbf{u}}^\top, \bar{e}_x]^\top$, $P = \begin{bmatrix} \varepsilon_u I & -\varepsilon_u \bar{C}^{-1}\hat{C}^\top \\ -\varepsilon_u \hat{C}\bar{C}^{-1} & (2\varepsilon_e + \varepsilon_u \hat{C}\bar{C}^{-1}\bar{C}^{-1}\hat{C}^\top) \end{bmatrix} \succ \mathbf{0}$, and $\bar{C} = (\hat{C}^\top \hat{C} + \mu I)$. We thus have $\lambda_{\min}(P)\|\boldsymbol{\rho}_u\|^2 \leq V_b(\bar{\mathbf{u}}, \bar{e}_x) \leq \lambda_{\max}(P)\|\boldsymbol{\rho}_u\|^2$, where $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ are the minimum and maximum eigenvalues of the matrix in the argument.

The time-derivative ($\frac{d}{d\tau_u}$) of V_b along the boundary layer system trajectories (Eq (14)) is

$$\begin{aligned} \frac{dV_b}{d\tau_u} &= \varepsilon_u(\bar{\mathbf{u}} - \mathbf{u}^*)^\top \dot{\bar{\mathbf{u}}} + 2\varepsilon_e \bar{e}_x \dot{\bar{e}_x}, \\ &= -(\bar{\mathbf{u}} - \mathbf{u}^*)^\top (\bar{C}\bar{\mathbf{u}} - \hat{C}^\top \bar{e}_x) - 2\bar{e}_x C\bar{\mathbf{u}}, \\ &= -\bar{\mathbf{u}}^\top \bar{C}\bar{\mathbf{u}} + \bar{\mathbf{u}}^\top \hat{C}^\top \bar{e}_x + \bar{e}_x \hat{C}\bar{C}^{-1}\bar{C}\bar{\mathbf{u}} - \bar{e}_x \hat{C}\bar{C}^{-1}\hat{C}^\top \bar{e}_x - 2\bar{e}_x C\bar{\mathbf{u}}, \\ &= -\bar{\mathbf{u}}^\top \bar{C}\bar{\mathbf{u}} - \bar{e}_x \hat{C}\bar{C}^{-1}\hat{C}^\top \bar{e}_x + \bar{e}_x(2\hat{C} - 2C)\bar{\mathbf{u}}, \\ &= -\boldsymbol{\rho}_u^\top Q \boldsymbol{\rho}_u + 2\bar{e}_x \tilde{C}\bar{\mathbf{u}}, \\ &\leq -\lambda_{\min}(Q)\|\boldsymbol{\rho}_u\|^2 + 2\|\bar{e}_x\| \|\tilde{C}\| \|\bar{\mathbf{u}}\|, \end{aligned} \quad (15)$$

where, $Q = \begin{bmatrix} \bar{C} & 0 \\ 0 & \hat{C}\bar{C}^{-1}\hat{C}^\top \end{bmatrix} \succ \mathbf{0}$. Using the fact that $\|\boldsymbol{\rho}_u\|^2 = \|\bar{\mathbf{u}}\|^2 + \|\bar{e}_x\|^2$, we get

$$\frac{dV_b}{d\tau_u} \leq -\lambda_{\min}(Q) \left(\|\bar{\mathbf{u}}\|^2 + \|\bar{e}_x\|^2 \right) + 2 \|\bar{e}_x\| \left\| \tilde{C} \right\| \|\bar{\mathbf{u}}\| = -\boldsymbol{\psi}(\bar{\mathbf{u}}, \bar{e}_x)^\top \Lambda \boldsymbol{\psi}(\bar{\mathbf{u}}, \bar{e}_x), \quad (16)$$

where $\boldsymbol{\psi}(\bar{\mathbf{u}}, \bar{e}_x) = [\|\bar{\mathbf{u}}\|, \|\bar{e}_x\|]^\top$, and $\Lambda = \begin{bmatrix} \lambda_{\min}(Q) & -\|\tilde{C}\| \\ -\|\tilde{C}\| & \lambda_{\min}(Q) \end{bmatrix}$.

Note that $\lambda_{\min}(Q) = \min\{\lambda_{\min}(\bar{C}), \hat{C}\bar{C}^{-1}\hat{C}^\top\}$, where $\lambda_{\min}(\bar{C}) = \mu$. Since $\hat{C} = C + \tilde{C}$, we have $\|C\| - \|\tilde{C}\| \leq \|\hat{C}\|$, which gives us

$$\begin{aligned} \hat{C}\bar{C}^{-1}\hat{C}^\top &= \hat{C}(\hat{C}^\top \hat{C} + \mu I)^{-1}\hat{C}^\top = \hat{C}\hat{C}^\top(\hat{C}\hat{C}^\top + \mu)^{-1} \\ &= \frac{\|\hat{C}\|^2}{\|\hat{C}\|^2 + \mu} \geq \frac{(\|C\| - \|\tilde{C}\|)^2}{(\|C\| - \|\tilde{C}\|)^2 + \mu} > 0. \end{aligned}$$

Since $\lambda_{\min}(Q) > 0$, and Λ is a symmetric matrix, it is positive definite if

$\|\tilde{C}\| < \lambda_{\min}(Q) = \min\left\{\mu, \frac{(\|C\| - \|\tilde{C}\|)^2}{(\|C\| - \|\tilde{C}\|)^2 + \mu}\right\}$. The second bound gives us

$\|\tilde{C}\| < \frac{(\|C\| - \|\tilde{C}\|)^2}{(\|C\| - \|\tilde{C}\|)^2 + \mu}$, which is always satisfied for the largest value θ_c that satisfies the cubic inequality

$$\theta_c^3 - \theta_c^2(2\|C\| + 1) + \theta_c(\|C\|^2 + 2\|C\| + \mu) - \|C\|^2 < 0. \quad (17)$$

Therefore, if $\|\tilde{C}\| \leq \min\{\mu, \theta_c\}$ then $\frac{dV_b}{d\tau_u}$ is negative definite, and the boundary layer system is globally exponentially stable. $\square$

Remark 2. Inequality (17) is satisfied for $\theta_c = 0$. By continuity of the cubic equation in θ_c , there exists a non-zero measure set where this inequality is satisfied.

2.3 Coupled forward-inverse motor learning dynamics

We now show the stability and convergence of our proposed HML model using singular perturbation arguments. Define $\Omega_c = \left\{(\tilde{C}, \bar{\mathbf{u}}, \bar{e}_x) \in \mathbb{R}^{1 \times m} \times \mathbb{R}^m \times \mathbb{R} \mid \|\tilde{C}\| \leq \delta_c\right\}$, where $\delta_c = \min\{\mu, \theta_c\}$, and θ_c is the largest value that satisfies the cubic inequality (Eq (17)).

Theorem 1 (Stability of the HML Model (Eq (10))). *In the absence of exploration noise $\boldsymbol{\xi}_u$, there exists sufficiently small ε^* such that for all $\gamma^2/\eta k_P \in (0, \varepsilon^*)$, Ω_c is positively invariant, and for all $(\tilde{C}(0), \bar{\mathbf{u}}(0), \bar{e}_x(0)) \in \Omega_c$, the trajectories of the proposed HML model in Eq 9 exponentially converge to the origin.*

Proof. From the proof of Lemma 1, we have

$\|\tilde{C}(t)\| \leq e^{-\gamma \lambda_{\min}(\delta \mathbf{q} \delta \mathbf{q}^\top) t} \|\tilde{C}(0)\| \leq \|\tilde{C}(0)\|$, implying $\|\tilde{C}(t)\|$ is non-increasing. Thus, $\|\tilde{C}(0)\| < \delta_c = \min\{\mu, \theta_c\}$ implies $\|\tilde{C}(t)\| < \delta_c$ for all $t > 0$. Therefore, from Lemma 2, the boundary layer system is globally exponentially stable. Using similar techniques as in [4, Theorem 11.4], the exponential stability of the system can be proved by constructing a composite Lyapunov function using the Lyapunov functions $V_r(\tilde{C})$ and $V_b(\bar{\mathbf{u}}, \bar{e}_x)$, for $(\tilde{C}, \bar{\mathbf{u}}, \bar{e}_x)$ starting in the set Ω_c . $\square$

Remark 3. We show exponential convergence of HML model states (Eq (10)) to the origin in the absence of exploration noise in Theorem 1. With the stochastic exploration noise ξ_u added to the model, using ideas similar to [5, Section 8.6], it can be shown that the HML model trajectories enter and stay in a ball of radius $O(|\sigma_u|)$ centered around the origin.

3 HML model-based investigation into motor learning behavior

3.1 Exploration versus exploitation trade-off

Apart from the inverse learning parameter η which had the most significant effect on the exploration versus exploitation trade-off, we also show how the exploration noise intensity affects this performance metric in Fig 1.

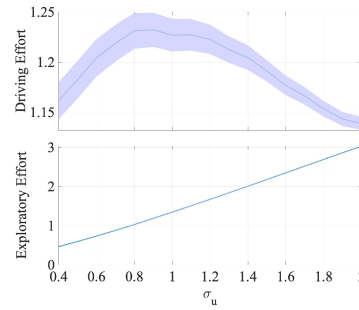


Fig 1. Effort variation with σ_u . Distribution of driving and exploratory effort across trials as σ_u is varied around its fit value 0.8764. Driving effort is highest around the fitted value of σ_u ($p < 0.01$), while exploratory effort increases monotonically with σ_u .

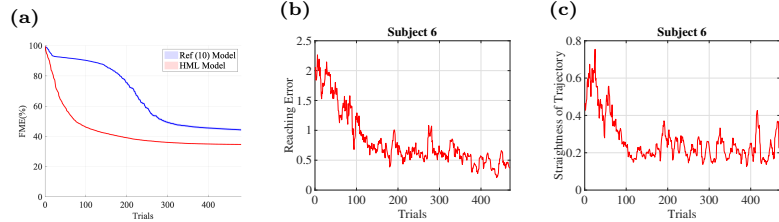


Fig 2. Comparing HML model with Ref [6] model. (a) Comparing the FME curves from the model in Ref [6] to the HML model shows that the model in Ref [6] is not as accurate as HML model in capturing the human skill state (represented by the participant's estimate of BoMI mapping matrix $\hat{C}$) for this motor learning task. (b) and (c) show the RE and SoT evolution curves for Subject 6.

3.2 Comparative analysis of HML model's efficacy in explaining the motor learning

Fig 2 shows the evolution of FME from the HML model and the model in Ref [6], both fitted on the experimental data of Subject 6. Looking at Fig 2, FME from our proposed model aligns more closely to the RE and SoT observed from the experimental data for Subject 6 as compared to the FME from the Ref [6] model. There's a sharp initial decrease in FME out of HML model Fig 2, and also in the RE and the SoT Fig 2c, 2b for

Subject 6. Whereas, FME out of the Ref [6] shows a slow initial decrease, followed by a fast decay much later in the game trials.

References

1. Sastry S, Bodson M. Adaptive Control: Stability, Convergence, and Robustness. Prentice-Hall, Inc.; 1989.
2. Ioannou PA, Sun J. Robust Adaptive Control. vol. 1. PTR Prentice-Hall Upper Saddle River, NJ; 1996.
3. Mareels IM, Gevers M. Persistency of excitation criteria for linear, multivariable, time-varying systems. Mathematics of Control, Signals and Systems. 1988;1:203–226.
4. Khalil HK. Nonlinear Systems. 3rd ed. Prentice Hall; 2002.
5. Kushner HJ. Introduction to Stochastic Control. Holt, Rinehart and Winston New York; 1971.
6. Pierella C, Casadio M, Mussa-Ivaldi FA, Solla SA. The dynamics of motor learning through the formation of internal models. PLoS Computational Biology. 2019;15(12):e1007118.